MIL-A-51027A
19 August 1968
SUPERSEDING
MIL-A-51027
27 June 1962

MILITARY SPECIFICATION

ANTISETTING COMPOUND, DECONTAMINATING SLURRY, M2

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of antisetting compound.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-390	-	Plastic Molding Material, Polyethylene, Low and Medium Density.
O-S-635	-	Sodium Polyphosphates, Technical.
RR-S-366	-	Sieve, Test.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-621	-	Boxes, Wood, Nailed, and Lock-Corner.
PPP-D-723	-	Drums, Fiber.
PPP-F-320	-	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.
PPP-T-60	-	Tape: Pressure-Sensitive Adhesive, Waterproof, for Packaging.

FSC 6850

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MIL-P-116	-	Preservation, Methods of.
MIL-B-131	-	Barrier Material, Water Vaporproof, Flexible, Heat-Sealable.
MIL-C-3955	-	Cans, Fiber, Spirally Wound.
MIL-C-12056	-	Calcium Oxide, Tehnical.
MIL-P-15011	-	Pallets, Material Handling, Wood, Post Construction, 4-Way Entry.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-147	-	Palletized and Containerized Unit Loads, 40 x 48" 4-Way (Partial) Pallet Skids, Runners, or Pallet-Type Base.

DRAWINGS

US ARMY MUNITIONS COMMAND

D150-2-1	-	Marking Diagram for Shipping Containers.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained frm the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

U.S. PHARMACOPOEIAL CONVENTION

Pharmacopoeia of the United States, Seventeenth Revision

(Copies of the U.S. Pharmacopoeia may be obtained from the Mack Publishing Company, Easton, Pennsylvania 18042.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM Standards

D1193-66 - Reagent Water.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Materials. Antisetting compound shall be manufactured from the following materials:

(a) Citric acid, anhydrous, fine granular, U.S.P. grade,
(b) Sodium tripolyphosphate, technical grade conforming to type III of O-S-635, and
(c) Calcium oxide, technical grade conforming to grade A of MIL-C-12056.

3.2 Chemical composition. Antisetting compound shall be a throughly blended mixture conforming to the characteristics of table I when tested as specified therein.

Table I. Chemical composition

Material	Percent by weight	Test paragraph
Citric acid, anhydrous	46.5 ± 1.0	4.2.4.1
Sodium tripolyphosphate	50.5 ± 1.0	4.2.4.2
Calcium oxide	3.0 ± 0.2	4.2.4.3

3.3 Particle size. A minimum of 95 percent by weight of the antisetting compound shall pass through a 4,760-micron (No. 4) sieve when tested as specified in 4.2.4.4.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the antisetting compound produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operations is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.2 Sampling.

4.2.2.1 For examination of preparation of delivery. Sampling for examination of preparation for delivery shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For test. Sampling for test shall be conducted in accordance with table II. A representative specimen of approximately 1 pound (lb) shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

Table II. Sampling for test.

Number of containers in batch or lot	Number of sample containers
2-25	2
26-150	3
151-1,200	5
1,201-7,000	8
7,001-20,000	10
Over 20,000	20

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4.2.3 Inspection procedure.

4.2.3.1 For examination of preparation for delivery. The sample unit shall be one filled unit or shipping container, as applicable, ready for shipment. Sample containers and the preparation for delivery thereof shall be examined for the following defects using an AQL of 4.0 percent defective:

(a) Contents per container not as specified
(b) Bag or container not as specified
(c) Bag liner not as specified
(d) Closure of bag or container not as specified
(e) Bag or container damaged (torn, broken, punctured, or crushed)
(f) Fiberboard sheets missing
(g) Plastic cup missing or damaged
(h) Marking incorrect, illegible, missing, or incomplete

4.2.3.2 For test. Approximately equal portions of all of the specimens except one taken in 4.2.2.2 shall be thoroughly mixed to form a composited specimen of no less than 1 lb. The composited specimen and the remaining individual specimen shall be tested separately and in duplicate as specified in 4.2.4, and the duplicates averaged. Failure of the average of any test by either the composited or the individual specimen to meet specified requirements shall be cause for rejection of the lot represented.

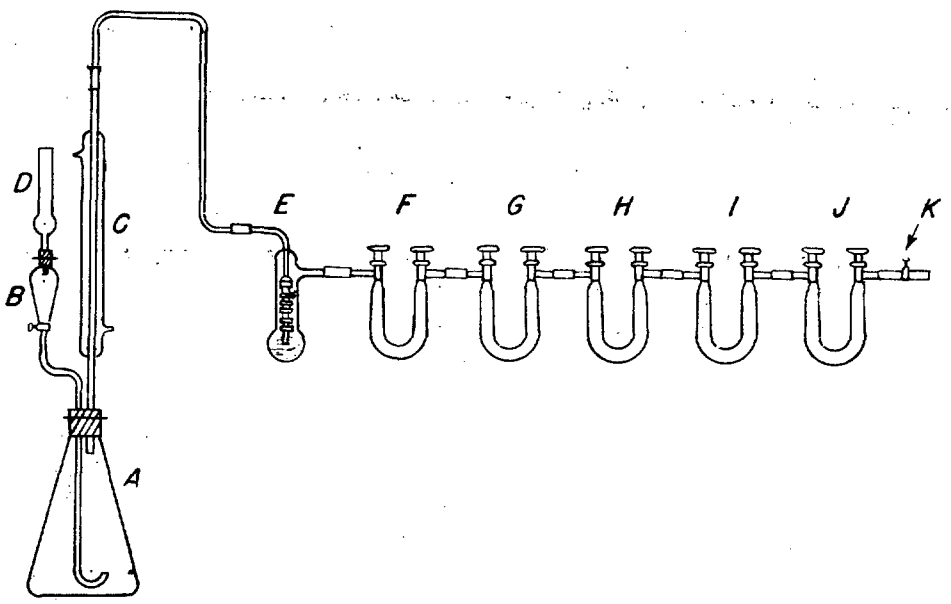
4.2.4 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

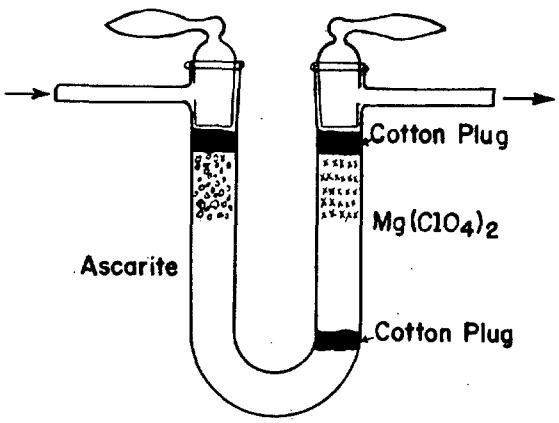
4.2.4.1 Citric acid.

(a) Potassium permanganate solution. Dissolve 48 grams (g) of potassium permanganate in one liter of water and heat the solution to boiling, keeping the solution slightly below the boiling point for one hour. Filter the solution through pure asbestos (free from organic matter) or through a sintered-glass filter crucible. Protect the solution from evaporation, dust, and reducing vapors. Keep the solution in the dark or in diffused light. If in time manganese dioxide settles out, refilter the solution.

(b) Apparatus. (See figure 1.) The decomposition vessel A is a 500-ml Erlenmeyer flask fitted with a two-hole rubber stopper through which pass a small condenser C and a 200- to 250-ml capacity dropping funnel B. The upper end of the dropping funnel is closed with a one-hole rubber stopper carrying a drying tube D filled with ascarite resting on a small plug of glasswool.

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DETAIL H-I

Fig 1 - Apparatus for direct determination of carbon dioxide

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The stem of the dropping funnel reaches nearly to the bottom of the flask and is bent upwards at its lower end to prevent entrance of carbon dioxide. The upper end of the condenser is connected to the purification train by means of short sections of fresh gum rubber or tygon tubing. In this, and other glass-to-glass connections, the two ends of the glass tubing should be in contact. The bottle E is partially filled with 1 to 1 sulfuric acid saturated with silver sulfate. The U-tube F is filled with glass beads and enough concentrated sulfuric acid to cover the bend in the bottom of the tube. The second U-tube, G, contains anhydrous magnesium perchlorate. The U-tubes H and I are the absorption tubes and are each filled two-thirds full of ascarite and one-third full of anhydrous magnesium perchlorate as shown in the insert on figure 1. The last tube J contains one-half anhydrous magnesium perchlorate and one-half ascarite. From the last U-tube a rubber hose leads to a flow regulating device K and to a vacuum source. All the U-tubes are hung by copper or brass wire from a rod and made as compact as possible.

(c) Procedure. First check the entire apparatus for leaks and make certain all connections are gas tight. When the train has been proved gas-tight, draw a current of air through it of about two or three bubbles per second through E for 15 minutes. Then close the stopcocks on U-tubes J, I, H, and G in the order given. Disconnect absorption tubes H and I and weigh them separately against a similar U-tube as a tare. Allow the tubes to stand in or near the balance case for one-half hour before weighing. Open a stopcock of each absorption tube and of the tare for an instant before weighing. Replace the absorption tubes in the train, open stopcocks on tubes G, H, I, and J in order given, and repeat operation of drawing air through the tubes until the weight is constant to 0.5 milligram (mg). When constant weight has been obtained, weigh 1.0 g of antisetting compound to the nearest 0.1 mg, dissolve in 75 ml of water, and add 6.0 ml of 1 to 1 sulfuric acid. With all stopcocks in the absorption train closed, disconnect flask A and rinse the acidified antisetting compound solution into the flask. Replace flask A. Disconnect G from H. Open stopcocks on F and G. Apply suction to the open end of G for five minutes. Reconnect tubes G and H. Close stopcock on B. Remove D. Add 200 ml of potassium permanganate solution, prepared as specified in (a), to B. Add enough water to displace any remaining air above the permanganate solution. Replace D. Open stopcocks H, I, and J. Open stopcock B and start air flow through the absorption system at about three bubbles per second through E. Start water flowing through condenser C. Apply heat to A gently until the solution boils; then reduce heat. Keep the solution just below boiling for 15 minutes; then remove heat. Continue air flow through the absorption train for at least 45 minutes after heat is removed. Close vacuum source K and

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stopcocks J, I, H, and G. Remove and reweigh absorption tubes H and I separately against the respective tare, taking the same precautions as in the first weighings. Calculate the percent citric acid as follows:

$$\text{Percent citric acid} = \frac{72.5\ A}{W}$$

where: A = Weight of carbon dioxide in grams and
W = Weight of specimen in grams.

4.2.4.2 Sodium tripolyphosphate.

(a) Ammonium molybdate solution. Dissolve 200 g of ammonium molybdate in a mixture of 800 ml of water and 160 ml of 27-percent ammonium hydroxide.

(b) Ammonium nitrate solution. Dissolve 55 g of ammonium nitrate in a mixture of 400 ml of nitric acid and 600 ml of water.

(c) Wash solution. Dissolve 10 g of ammonium nitrate and 5 ml of nitric acid per liter of water.

(d) Preparation of specimens. Crush 50 to 60 g of the specimen fine enough to pass an 840-micron (No. 20) sieve. Weigh to the nearest 0.01 g approximately 15 g of the crushed specimen, dissolve in water, and dilute to 1 liter. Transfer 100 ml of this solution to another liter flask and dilute to volume.

(e) Procedure. Transfer a 50-ml aliquot of the solution prepared as specified in (d) to a 600-ml beaker containing 100 ml of 1 to 1 nitric acid. Cover with a watch glass and boil gently for 30 minutes. Cool to room temperature. To another 600-ml beaker, add 90 ml of ammonium nitrate solution prepared as specified in (b) and 45 ml of ammonium molybdate solution prepared as specified in (a). To this solution, maintained at 25° ± 1° C, add, with constant stirring, the treated aliquot of specimen solution. Allow the material to stand at 25° ± 1° C for 2 hours with occasional stirring. Filter, using a tared Gooch crucible weighed to the nearest 0.1 mg and wash with the wash solution prepared as specified in (c). Dry the precipitate for 3 hours at 150° ± 2° C and weigh to the nearest 0.1 mg. Calculate the percent sodium tripolyphosphate as follows:

$$\text{Percent sodium tripolyphosphate} = \frac{6.73A}{W}$$

where: A = Weight of precipitate in grams and
W = Weight of specimen in the 50-ml aliquot in grams.

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4.2.4.3 Calcium oxide. Weigh 6.0 g of the specimen to the nearest 0.1 mg into a 600-ml beaker. Add 100 ml of concentrated nitric acid and boiling chips (glass beads or Berl saddles). Boil gently until oxidation ceases as evidenced by the evolution of white fumes. Add additional nitric acid if necessary. Do not allow the contents of the beaker to go to dryness. Dilute the contents of the beaker to 100 ml and add several drops of methyl red indicator. Add concentrated ammonium hydroxide until within a few drops of the color change. Dissolve 1.5 g of sodium oxalate in 200 ml of water, heating if necessary. Add the sodium oxalate solution to the beaker, stir well, and add concentrated ammonium hydroxide dropwise until the indicator changes to yellow. Bring the contents of the beaker to a boil to aid the formation of calcium oxalate. Cool and filter through paper suitable for filtering fine precipitates. Wash the precipitate thoroughly with water. Place a 500-ml iodine flask under the funnel and dissolve the calcium oxalate with hot 1.5N sulfuric acid. Heat the flask contents to near boiling and titrate with 0.1N potassium permanganate solution. Calculate the percent calcium oxide as follows:

$$\text{Percent calcium oxide} = \frac{2.8\ AB}{W}$$

where: A = Milliliters of potassium permanganate solution used,
B = Normality of potassium permanganate solution, and
W = Weight of specimen in grams.

4.2.4.4 Granulation. Place an accurately weighed specimen of 100.0 g on the specified sieve conforming to RR-S-366 and assemble with a bottom pan. Cover and shake for 3 minutes by hand or by means of a mechanical shaker geared to produce 300 ± 15 gyrations and 150 ± 10 taps of a striker per minute. Weigh the portion passed by the sieve and calculate the percent passing.

5. PREPARATION FOR DELIVERY

5.1 Packaging, level A. Twelve and one-half lb ± 2 ounces of antisetting compound shall be packaged in a heat-sealable bag fabricated from barrier material conforming to class 1 of MIL-B-131 except that the heat-sealable ply in contact with the contents shall be restricted to polyethylene. All seams, including that of the closure of the filled bag, shall be heat sealed and shall meet the heat-seal seam requirements of MIL-P-116. In addition, the sealed filled bag shall not leak when tested for leakage in accordance with submethod IA-8 of MIL-P-116. The filled bag shall then be placed in either a fiber can or drum as specified (see 6.2). One plastic 3-ounce measuring cup, fabricated from polyethylene conforming to type I, class M, grade 3, natural color of L-P-390 shall be placed on top of the filled bag of antisetting compound.

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The fiber can shall conform to type II, grade B, style b, class 3 of MIL-C-3955. The fiber drum shall conform to type III, grade A of PPP-D-723. The fiber drum or can shall have an inside diameter of 7-1/8 ± 1/8 inch and a height of 12 ± 1/8 inch. The seam of the can shall be completely sealed by tape of 3-inch minimum width conforming to class 3 of PPP-T-60. Closure of the drum shall be as specified in PPP-D-723.

5.2 Packing, level A. Four cans or drums of antisetting compound as specified (see 6.2), packaged as specified in 5.1, shall be packed for shipment in a nailed wood or cleated plywood box as specified (see 6.2). The nailed wood box shall conform to class 2, style 4-1/2, for a type 2 average load of PPP-B-621. The cleated plywood box shall conform to type overseas, style A or B, for a type 2 average load of PPP-B-601. Half-slotted fiberboard sheets of full box length, width, and height, shall be used to form cells in the plywood or fiberboard box. In addition, all inside faces of each box shall be lined by fiberboard sheets in sufficient quantity to assure a tight pack. All fiberboard sheets shall conform to grade W5c of PPP-F-320. Each box shall be closed and reinforced in accordance with the closure and reinforcing requirements in the applicable box specification. Tape banding shall be used to reinforce the fiberboard box.

5.3 Marking. In addiiton to any special marking required by the contract or order, unit packages shall be marked in accordance with MIL-STD-129 and shipping containers shall be marked as shown on US Army Munitions Command Drawing D150-2-1 (see 6.2).

5.4 Palletization. When specified in the contract or order, shipping containers shall be palletized in accordance with the applicable requirements of MIL-STD-147 except that the strapping shall be restricted to galvanized steel of 1.25 inch minimum width and 0.035 inch minimum thickness and the tie-down straps parallel to the 40-inch dimension only of the pallet of MIL-P-15011 shall be used. Horizontal strapping shall be applied to each tier of boxes. Any wood cleats or storage aides, as well as the pallet, shall be preserved as specified for the pallet of MIL-P-15011 (see 6.2).

6. NOTES

6.1 Intended use. Antisetting compound, decontaminating slurry, M2 is intended for use in retarding settling and setting in decontaminating slurry mixtures.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of fiber packaging container required.
(c) Type of shipping container required.
(d) Special marking other than 5.3, if required.
(e) Palletization, if required.

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process and subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value shall be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of the Recommended Practices for Designating Significant Places in Specified Limiting Values (ASTM E29).

Custodians:

Army - MU
Navy - YD

Preparing activity:

Army - MU(EA)

Review activities:

Army - MD
Navy - YD
DSA - GS

Project No. 6850-0241

User activities:

Army - SM
Navy - MC

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